

CatBoost: From First Principles

CatBoost is a gradient boosting algorithm designed to work especially well with categorical features.

Why CatBoost Is Needed

Many real datasets contain categories such as city, product type, browser, or occupation. Handling these categories poorly can hurt model performance.

Categorical Features

CatBoost uses special techniques to encode categorical features while reducing target leakage and overfitting.

Ordered Boosting

CatBoost uses ordered boosting ideas to avoid using information from the target in a way that leaks future knowledge into training.

Core Idea

Like other boosting methods, CatBoost builds trees sequentially. Each new tree improves the current ensemble.

When to Use It

Use CatBoost when your tabular data has important categorical features. It can work very well with less manual preprocessing than many other boosting tools.